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1 Introduction

Everyday, there are millions of passengers worldwide who experience the process of airplane boarding. Airlines have adopted a variety of boarding strategies in the hope of reducing the gate turnaround time for airplanes (Usually turnaround time (or turn time) is measured by the time between an airplane's arrival and its departure). Significant reductions in gate delays would improve on the quality of life for long-suffering air travelers, and yield significant economic benefits from more efficient use of aircraft and airport infrastructure; (see van den Briel et al. [1], van Landeghem and Beuselinck [2], Marelli et al. [3]). So recently, commercial airlines have paid a great deal of attention to turnaround time. One of the main factors cited for the success of discount (or no-frills) airlines is the quick turnaround of their airplanes, which helps them achieve high airplane utilization (Allen 2000, Michaels 2003). So, they make efficient use of their primary capital investment, the aircraft. Long turnaround times decrease revenue-producing flying time, while short turnaround times please customers and can increase airlines' revenues.

What is the fastest way to load people into an airplane? Certainly not the way that it is normally done, as any business traveler will attest. We are all too familiar with the inefficiencies of boarding: You dutifully line up on time, but then get backed up by some family that blocks the entire aisle while sitting down; then as soon as you are finally secure in your aisle seat, your window seat-mate arrives and you have to stand up again. This is why we need to devise a new procedure for boarding and deboarding planes to reduce the gate turnaround time.

In general, there are about five common passenger boarding strategies:

(1) The basic back-to-front procedure is probably the most familiar, and it is still used by many airlines. It board passengers from the back of the airplane first.

(2) There is the outside-in strategies, nicknamed Wilma, for window first, then middle and then aisle, a strategies favored by Delta and United.

(3) The airline with the single-fastest boarding time is Southwest Airlines -- which does not have any seat assignments at all. The passengers board in the order they arrive.

(4) There is one of most complex procedures-- the reverse pyramid system. It is basically Wilma, with seats filled in a pattern as intricate as a microchip's circuitry: rear window and middle first, front window and middle next, followed by rear aisle, then front aisle.

(5) The last method is called rotating zone system. It divides the plane into six different sections. Then the airline begins the seating sequence with the back five rows called first, the front five behind business class second, the next back five rows third — continuing until the rows meet in the middle.

In this paper, our task is to devise and to compare procedures for boarding and deplaning with varying numbers of passengers: small (85–210), midsize (210–330), and large (450–800). Then prepare an executive summary, not to exceed two single-spaced pages.

2 Symbols and Terms

Table 1
Symbols

x_i	the i th passenger in the boarding queue
σ_t	The intervals between successive passengers
a_{ij}	The passenger's seat is at the i th row and the j th seat
T_{ij}	the passenger's time spend on achieving boarding
T_s	the time for putting the baggage
T	the total time for boarding need
RI	the aisle interference degree
SI	seat interference degree
CD	chaos degree
T_{RI}	the delay period for aisle interference
T_{SI}	the delay period for seat interference
λ_{ij}	the priority for boarding to the passengers whose seat is at a_{ij}
t_i	time needed from the gate to reach the i th row
t_j	time needed from the same row to reach the j th seat
P_{RI}	probability for aisle interference happening
P_{SI}	probability for seat interference happening
OD	Order-degree
u	the utilization of the aisle

3 Problem Analysis

If we want to devise a faster way to load passengers in the aircraft, we must analyze the cause of gate delays and eliminate these disadvantages as far as possible. So as a Chinese proverb goes, 'Knows oneself and the other side, being undefeated in many battles', what we need to do first is to sort out the boarding process.

The actual boarding process consists of three steps [4]:

1. Passengers are called by the gate agent, announcing the start of boarding. Passengers will typically start queuing at the gate, with occasional shoving and confusion as to the position and direction of the queue. Delaying factors are missing passengers, that need to be paged, or late arrival of transfer partners. The increase of the hub-and-spoke scheduling of flights has increased the occurrence of this type of delay considerably.

2. The gate agent controls the boarding process of each passenger, and registers his entry using a ticket reader. This step in the procedure is the only moment we have control on who enters at what time. The main disturbances are jamming of the card reader, and the occasional withdrawal of excessive carry-on luggage at the gate entrance (with sometimes prolonged discussions with the affected passenger). As airlines tighten their control on carry-on luggage limits, this kind of delay is also increasing.

3. Finally passengers access the airplane through the access bridge and the front door. When passengers are taken by bus to the airplane, the same process holds, but now passengers arrive at the airplane door in large batches, which only aggravates the problem. Once inside the plane, they have to queue in a single line (due to the small aisle width) until they reach their assigned seat. Several disturbances and delays occur within the plane. Finally, the passenger puts his carry-on luggage either in the overhead compartment (bin) or underneath the seat in front of him, sits down, and affixes his security belt.

From the analysis of boarding process, it can be concluded that the second step is the preparatory stage to the boarding and it would affect the efficiency of the passengers' boarding indirectly. But step 1 and 2 do not contribute to the turnaround time and if the carry-on luggage limits has determined, it is necessary to be pay more attention to the problems such as how many batches the all passengers are separated and what is the order of boarding process for each groups. In this paper, what we more care about is how to decrease the interference between rows and seats by arranging a better sequence of boarding. So we divide the interference into two types: aisle interference and seat interference. Obviously if the groups are ordered from back to front, the congestion of the aircraft aisle will be minimized by freeing the journey of the passengers to the back of the aircraft from aisle obstacles. In a similar way, if the groups' boarding sequence is started from the window side to aisle side, the seat interference can be reduced.

But how we can determine which method is better? Considering each passenger's spending on boarding changes along with the distance of their own seats and the aisle jamming situation, but we did not know the explicit numerical value about the speed of theirs', and the time to take to put the luggage etc. So we introduce a new variable called "chaos degree" that is to be denote the value of interference.

4 Assumptions

- There are many factors affecting the boarding time. In order to simplify the problem, we assume that all passengers arrive in time.
- Though every passenger's speed is different, in our model we ignore this difference.
- We assume the time for putting away the luggage is equal.
- The first passenger passes through the gate, announcing the start of boarding. The people aboard plane one by one. Total time covers the period from the first passenger pass from the gate to the last one.

- Almost all procedures arrange the special groups (usually special needs passenger and first class) boarding first, so we omit the cost of their boarding and only to pay attention to the coach and business-class passengers.
- It takes longer for a traveler to store more pieces of luggage, so in order to simplify the problem, we restrict passengers are all carrying one piece.
- We assume that passengers are assigned seats in the airport in advance of the boarding process.
- The boarding position of a particular passenger within the group is equal probability.

5 The Model

5.1 The ideal situation:

It is assumed that the aisle is wide enough, so there is not any seat interference and aisle interference when passenger boarding. If we use a_{ij} to denote the passenger whose assigned seat is on the i th row and on the j th grade, T_{ij} denotes the passenger's time spend on achieving boarding. Then T_0 is defined to be the start time. $T_0 = 0$. The T_{ij} can be separated to two parts: the one is the time take to pass the aisle, which is denoted by t_i , and the other is time for putting away the baggage, which is denoted by T_s . The total time of boarding is determined by the last boarded passenger.

If we assume the intervals between the two conterminous passengers is σ_t , there are N passengers, and the k th passenger is the last one to reach his/her seat, and the passenger's assigned seat is on i th row and j th seat. since there is no interference, we obtain that:

$$T = (k-1)\sigma_t + T_{ij} \quad (T \text{ Express the total time needed for boarding}).$$

To reduce the total boarding time what we should do is to give the priority for boarding the passengers whose seat is at the back of plane. In this way, it can be ensured that the last boarding passenger is the last boarding one.

The realistic situation

In fact, the aisle is narrow and there is only one passenger can pass through at a time. Once one passenger arrives at his designated seat, it takes him a fixed amount of time to stow his luggage and clear the aisle and sit to his right seat. Then the interference will occur, and it will cause passengers to wait. We define two types of the interferences: seat interference and aisle interference. Seat interference occurs when

passengers seated close to the aisle block other passengers seated in the same row. Aisle interference occurs when passengers stowing luggage in overhead bins block other passengers' access to seats. We use symbol RI and SI to denote the aisle interference and seat interference degree. The interference degree is the number of jamming. Since these types of interference will arouse different time delay, we weight them with ω_r , ω_s and then introduce a new variable called CD ('chaos degree') to denote the comprehensive evaluation of the two types of congestion.

$$CD = \omega_r \cdot RI + \omega_s \cdot SI$$

◆ The comparison model

Since it has been assumed that all the passengers are divided into several batches, we define a discrete process which models the airplane boarding process. Then we assume that passengers are assigned seats in the airplane in advance of the boarding process. The input data which determines an instance of the boarding process is composed of the following items:

(1) A sequence of passengers' $x_1 \dots x_n$, where x_i denotes the i th passenger in the boarding queue. Each passenger x_i has a seat in an assigned row, denoted by $r_i = r(x_i)$.

(2) A delay value $d_i = d(x_i)$, which measures the amount of time it takes from the moment passenger x_i has reached his designated row until he clears the aisle: $d_i = d(x_i) = T_{ij}$. This time includes proceeding (this is denoted by t_i), passing by previously seated passengers from the same row on the way to the designated seat (this is denoted by t_j) and, possibly, storing carry-on luggage (this is denoted by T_s).

(3) If the middle or aisle seat has been taken when the window or middle seat mate arrives, it usually requires the seated passengers to get up and sit back after the newly arrived passenger has taken his/her seat. We assume the different situation of reaching the seat may cause a different interference degree. So it needs to be determined the values for the penalties for different situation. The procedure that is followed in our paper is based on probabilistic expectations. The main assumption in the determination of these expectations is that the boarding position of a particular passenger within the group is equally likely. That is, a particular passenger within a group can take any of different boarding positions within that group, with the same

probability. Based on these assumptions, the expected number of interference will be computed and used as the penalty represented in the objective function of the model by the $SI^{(i)}$. [5]

Table 1. Expected seat interference

Penalty	Passenger order	E(#Interferences)
$SI^{(1)}$	$[window] \rightarrow [aisle] \rightarrow [middle]$	1
$SI^{(2)}$	$[middle] \rightarrow [window] \rightarrow [aisle]$	1
$SI^{(3)}$	$[middle] \rightarrow [aisle] \rightarrow [window]$	2
$SI^{(4)}$	$[aisle] \rightarrow [window] \rightarrow [middle]$	2
$SI^{(5)}$	$[aisle] \rightarrow [middle] \rightarrow [window]$	3

Given the above input parameters, the boarding process is described as follows: Passengers go along the aisle as far as they can, either reaching their assigned row or lining up behind other passengers, who are either getting organized during the delay period or are themselves lined up behind other blocking passengers. As the analysis above, when the passenger is stowing luggage, he will block some passengers sit into to their seats. Then the aisle interference occurs. So it will take the following passenger T_{RI} to wait. In a similar way, it will make the successive passenger T_{SI} wait if passengers sitting close to the aisle block other passengers sitting in the same row. So

$$T = (k-1)\sigma_t + T_{ij} + \sum T_{RI} + \sum_{i=1}^5 T_{SI^{(i)}}$$

Assuming the k th passenger is the last one to reach his/her designate seat, and the symbol T_{RI} T_{SI} represent the delay period for interference.

Obviously,

$$\Delta(\sum T_{RI} + \sum T_{SI}) \gg \Delta((l-1)\sigma_t + T_{ij})$$

We can conclude that the number of the interference is the main cause for affecting turnaround time. Thus we construct a comprehensive evaluation model:

The evaluation function is:

$$CD = \omega_r \cdot RI + \omega_s \cdot SI$$

ω_r and ω_s is determined by the degree of two types of interference. And we can use

computer to simulate the boarding process to get the expected degree of interference.

◆ Simulation

If the 138 seats in serial numbers are coded beforehand, we can simulate the order of passengers' boarding by producing a stochastic sorting of these numbers, then we can obtain the corresponding chaos degree, e.g.

10	20	30		25	15	5
9	19	29		24	14	4
8	18	28		23	13	3
7	17	27		22	12	2
6	16	26		21	11	1

Fig 1

(1) Back-to-front

The back-to-front approach will create congestion in a reduced area of the aisle among passengers in the same group, impeding their access to overhead bins for stowing carryon luggage and making it difficult for them to reach their assigned seats. In this procedure, passengers are separated five groups, and there is no interval between successive groups. We code the serial numbers to each passenger of each group, e.g.

122	130	138		134	126	118
121	129	137		133	125	117
120	128	136		132	124	116
119	127	135		131	123	115
98	106	114		110	102	94
97	105	113		109	101	93
96	104	112		108	100	92
95	103	111		107	99	91
70	80	90		85	75	65
69	79	89		84	74	64
68	78	88		83	73	63
67	77	87		82	72	62
66	76	86		81	71	61
40	50	60		55	45	35
39	49	59		54	44	34
38	48	58		53	43	33
37	47	57		52	42	32
36	46	56		51	41	31
10	20	30		25	15	5
9	19	29		24	14	4
8	18	28		23	13	3

7	17	27		22	12	2
6	16	26		21	11	1

Fig 2

Then we create a stochastic sorting for these numbers by computer. In this way, we can simulate the order of the passengers for entering the airplane. Thus we can get the interference degree about this procedure.

In a similar way, we can simulate the other three strategies. The difference between these methods only lies in the initial random sequence.

(2) rotating zone system

It first seats the back five rows of the plane, then the front five, and continues rotating back-front-back until boarding is complete, e.g.

40	50	60		55	45	35
39	49	59		54	44	34
38	48	58		53	43	33
37	47	57		52	42	32
36	46	56		51	41	31
98	106	114		110	102	94
97	105	113		109	101	93
96	104	112		108	100	92
95	103	111		107	99	91
122	130	138		134	126	118
121	129	137		133	125	117
120	128	136		132	124	116
119	127	135		131	123	115
70	80	90		85	75	65
69	79	89		84	74	64
68	78	88		83	73	63
67	77	87		82	72	62
66	76	86		81	71	61
10	20	30		25	15	5
9	19	29		24	14	4
8	18	28		23	13	3
7	17	27		22	12	2
6	16	26		21	11	1

Fig 3

(3) outside-in

This outside-in system boards all window passengers first, followed by those with middle seats and, finally, those seated in the aisle, e.g.

46	92	138		115	69	23
45	91	137		114	68	22

44	90	136		113	67	21
43	89	135		112	66	20
42	88	134		111	65	19
41	87	133		110	64	18
40	86	132		109	63	17
39	85	131		108	62	16
38	84	130		107	61	15
37	83	129		106	60	14
36	82	128		105	59	13
35	81	127		104	58	12
34	80	126		103	57	11
33	79	125		102	56	10
32	78	124		101	55	9
31	77	123		100	54	8
30	76	122		99	53	7
29	75	121		98	52	6
28	74	120		97	51	5
27	73	119		96	50	4
26	72	118		95	49	3
25	71	117		94	48	2
24	70	116		93	47	1

Fig 4

(4) the reverse pyramid system

It is basically Wilma, with seats filled in a pattern as intricate as a microchip's circuitry: rear window and middle first, front window and middle next, followed by rear aisle, then front aisle, e.g.

84	112	138		125	98	70
83	111	137		124	97	69
82	110	136		123	96	68
81	109	135		122	95	67
56	80	134		121	66	42
55	79	133		120	65	41
54	78	132		119	64	40
53	77	131		118	63	39
52	76	130		117	62	38
28	75	129		116	61	14
27	74	128		115	60	13
26	73	127		114	59	12
25	72	126		113	58	11
24	71	108		94	57	10
23	51	107		93	37	9
22	50	106		92	36	8

21	49	105		91	35	7
20	48	104		90	34	6
19	47	103		89	33	5
18	46	102		88	32	4
17	45	101		87	31	3
16	44	100		86	30	2
15	43	99		85	29	1

Fig 5

◆ Result analysis

From the result of the simulations (see in the appendix, program and the results), the outside-in is the best method, second is the reverse pyramid system, the third is back to front, and the last is rotating zone system:

$$CD_{rotating-zone} > CD_{back-to-front} > CD_{reverse-pyramid} > CD_{outside-in}$$

As a matter of fact, the thought of these boarding procedures is same. All is to let the passengers enter by grouping. The essence of this thought is giving passengers different priority for boarding which is based on the location of the seats. Let λ_{ij}

denote the priority for boarding to the passengers whose seat is at a_{ij}

As what we have analyzed, the back seat and the window side seat should be given higher priority. The back first can reduce the aisle interference, and the window side first can reduce the seat interference. We can get the following conclusion through analyzing the cause of interference:

✧ The aisle interference :

It is only affected by the number of the groups, and the more the passengers get in every batch, the bigger its probability for occurring. It is denoted by P_{RI} .

✧ The seat interference:

The more the passengers who want to reach the same rows and same side they belong to, the bigger probability for seat interference is. It is denoted by P_{SI} .

✧ Back to front:

In this procedure, there are so many passengers whose seat is on the same side of the same rows that aggravate the seat interference in each group. And it leads to the augmentation of the expected chaos degree.

$$P_{SI} \uparrow \Rightarrow SI \uparrow \Rightarrow CD \uparrow$$

✧ Rotating zone system:

This strategy is based on 'back to front', so its probability for seat interference is also great. Further more, this method may aggravate the seat interference at the gate, and it would decrease the utilization for the aisle, so its effect may be worse than the above procedure.

✧ Outside-in:

Since this method ensures the passengers are all on the different sides of rows, it avoids the happening for seat interference. There is only aisle interference may occur when passengers enter in, so it may be better than the above two procedures. But the worst situation for this method may lead to the longest turn time in these four methods.

$$P_{RI} \uparrow \Rightarrow RI \uparrow \Rightarrow CD \uparrow$$

✧ The reverse pyramid system:

It is based on the 'outside-in', they have a similar principle that avoids the seat interference, so there is also only aisle interference. But compared with the 'outside-in', it have a greater probability for the front congestion, and it would decrease the utilization of aisle, so the effect for this method may a little less than the 'Outside-in'. But on the other hand, the worst situation for it may less than the 'Outside-in'. So its comprehensive effect may be better than 'Outside-in'.

This is why the outside-in system is relatively better than other three strategies.

The step to separate the passengers to several groups is rational, but the passengers are all self-organized when they enter by batch. So it is not rational. Obviously, the order-degree is determined by the number of batches. We use OD to denote the order-degree, and use Num_{group} denote the number of groups

$$OD = \theta \cdot Num_{group}$$

Is the order-degree affecting the chaos degree? We increase the groups, and estimate the chaos degree by simulating, then obtain a result (see in the appendix). The more the groups are separated, the less the passengers are in each group, and it may increase the order degree. From the result, we found if we increase the batches, the effect for 'outside-in' and 'rotating zone system' may decline, but the effect for the other two may be better. What is the reason for this phenomenon? In fact, the happening for this phenomenon is not the enlargement of the order-degree, but the different impact from two type interference.

For 'back to front' and 'rotating zone system', increasing the groups may augment the probability for seat interference:

$$OD_{rotating-zone} \uparrow \& OD_{back-to-front} \uparrow \Rightarrow P_{SI} \uparrow$$

But for 'outside-in' and 'the reserve pyramid system', it will decrease the passengers in each group, and it will decrease the probability for aisle interference.

$$OD_{reverse-pyramid} \uparrow \& OD_{outside-in} \uparrow \Rightarrow P_{RI} \downarrow$$

So the chaos degree can be reduced, if we cut down the aisle interference by increasing the order-degree.

◆ New procedures

From the above analysis, we can get the conclusions that:

- 1) The expected aisle interference will be increase while the expected seat

interference improves, and it leads to that the impact of seat interference on chaos degree is more than the aisle interference. Thus we must avoid the seat interference as far as possible. It is necessary to ensure the passengers are not on the same side of the same row in each group.

- 2) The expected aisle interference is determined by the passengers of each group, so we may need to increase the number of groups, and make the number of the passengers be approximately equal.
- 3) To reduce the probability for gate congestion can improve the utilization for aisle, so we may ensure to seat the back of plane first.

According to the above analysis, we devise two procedures:

A. Improved Outside-in Strategy:

The disadvantage of 'outside-in' is that there are so many passengers in each batch that the expected aisle interference enlarges. Therefore, we improve the order-degree by separating more groups. In this way, the aisle congestions may be reduced, and it would decrease the chaos degree, e.g.

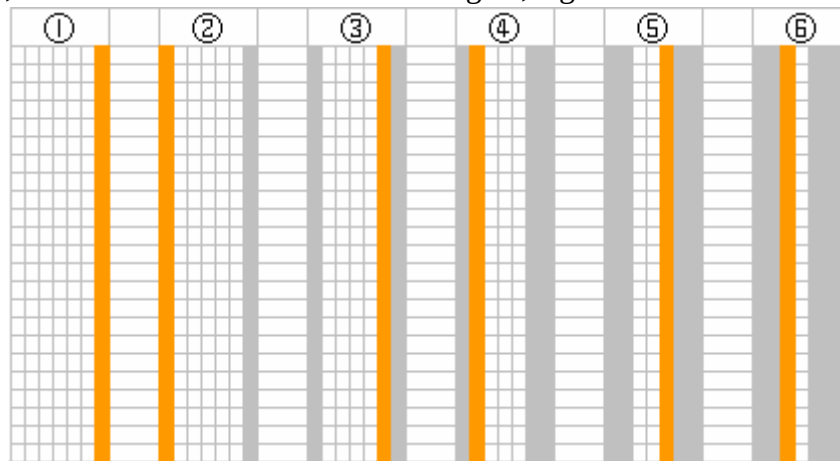


Fig 6

Considering the realistic situation, the excessive groups may take the gate agent more time to achieve the preparation work, and it may complicate the task. So it is necessary to determine a limitation for grouping. We determine the limitation to be 5 through the analysis for the actual situation. Then we devised a new procedure.

B. L-Shaped Procedure:

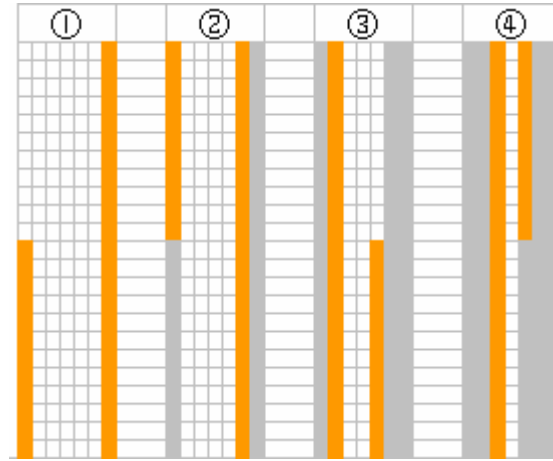


Fig 7

Simulating and comparing with the ‘outside-in’ and ‘the reverse pyramid system’ (the program is in appendix):

Table 2

Boarding procedure	groups	passengers	chaos degree
OUTSIDE-IN	3	46	42.054
REVERSE PYRAMID	5	28 (the last group has 26 passengers)	42.228
Improvement for 'outside-in' L-shaped sequence system	6	23	41.202
	4	35 or 34	41.76

$$CD_{\text{REVERSE PYRAMID}} > CD_{\text{OUTSIDE-IN}} > CD_{\text{L-shaped sequence system}} > CD_{\text{Improvement for 'outside-in'}}$$

Obviously, the improvement for ‘outside-in’ is the relatively better than other, but considering the limitation for grouping, the L-shaped sequence system is relatively better than the other three.

◆ The estimate model

We have got the comprehensive evaluation model, but now, we want to construct an estimate model to reckon average boarding time. Based on that we have assumed, we can obtain the follow function:

$$T = (k-1)\sigma_t + T_{ij} + \sum T_{RI} + \sum_{i=1}^5 T_{SI^{(i)}}$$

As we have got the aisle interference degree and the seat interference degree, we assume that the delay period of jamming is linear with the interference degree:

$$\sum T_{RI} = \mu \cdot RI$$

$$\sum_{i=1}^5 T_{SI^{(i)}} = \sum_{i=1}^5 \eta^{(i)} \cdot SI^{(i)} = \eta \cdot SI$$

Then we construct the object function:

$$T = (k-1)\sigma_t + T_{ij} + \mu_1 \cdot RI + \eta \cdot SI$$

From the above analysis, we may conclude that to reduce the number of the interference is the main cause for affecting turnaround time. So we assume C is a constant. ($C = (k-1)\sigma_t + T_{ij}$)

We utilize the least squares method to fit the function, i.e.

$$T = \mu \cdot RI + \eta \cdot SI + C,$$

based on which to evaluate the average boarding time of different procedures. (the program see in the appendix).

$$\mu = 0.2297 \quad \eta = 0.001 \quad C = 6.8401$$

The result:

Table 3

	ROTATING-ZONE		BACK-TO-FRONT		OUTSIDE-IN		REVERSE-PYRAMID		Improvement for 'outside-in'		L-shaped sequence system	
	RI	SI	RI	SI	RI	SI	RI	SI	RI	SI	RI	SI
Average boarding time(simulate 100 times)	80.76	67.77	78.55	69.64	70.13	0	70.79	0	68.14	0	69.46	0
cost (minutes)	25.4		24.9		22.9		23.1		22.49		22.759	

◆ Step model

The step-by-step stimulate model

In a sense, this model is the mirror of the simulation model. We present this model to compare with the previous one. In this model, we develop 'the utilization of the aisle' to substitute 'chaos degree'. We consider the aisle is too narrow to accommodate two people at the same time. Thus we regard the aisle as 23 successive grids, and then construct a state set for each grid: $S = \{0, 1, 2, 3\}$

Each lattice four states:

- (1) There is no one on it which is denoted by 0
- (2) 1 denote the state that there is someone who is in it but will go if possible

- (3) 2 represent the state that the row corresponding to the grid is the right number for the passenger's seat and he is putting away his luggage
- (4) 3 is the symbol by the state that the one is prepared to go to the seat which is assigned to him after State 2

Based on the assumption and the model, we institute three processes' the rule as below:

We execute it according to three processes: the exchange of the state, judgement of the movement and execution, the judgement whether let the passenger who is outside in.

- (1) The main role of the exchange of the state is modified the state which is altered after each step. The State 1 which is existed in the previous step or exchanged in Progress Three will change to State 2 if the row number corresponding to the grid that is the right row number to the passenger's seat who is just in the grid. Another situation is the State 2 which exist in the previous time will exchange to the State '3' when the luggage taking by the passenger has already been put away. The last situation is the State '3' will change to the State '0' if no obstruct occur or the occur is eliminated based on the consumption of time.
- (2) Judgement of the movement and execution is the main process, which is the promoting to the next step. We will judge every adjacent grid from the back row to the front row after Progress One. If the any two adjacent grids is in the situation that the relative back grid is in State '0' and the relative front grid is in State 1, then the passenger who is in the relative front grid will go forward one grid in this step.
- (3) The last process which is Process Three will decide the entering of the outside passenger. If the grid corresponding to the first row is found in State 0, the next passenger is allowed to go inside and the next thing to do is to exchange the State '0' to the State 1.

Then repeat the three processes until every grid is in the State 0.

Based on the theory, we make an easy way to calculate the step number which can use as the criterion to measure these five strategies. Take 10 sequences of the passengers from these five strategies as the sample stochastically and consider the average number of the sample as the expect step number of the strategies. The comparative table with these five strategies' expects step number in is showed as follows:

Table 4

strategy	OUTSIDE -IN	REVERSE PYRAMID	BACK-TO-F RONT	SOUTHWEST	ROTATING ZONE SYSTEM
Expectation step number	295	300	403	311	394

The program can see in the appendix.

In fact, the number of steps for stowing baggage is a constant. An optimal procedure can only reduce the steps for waiting to increase the efficiency of boarding. We use u to denote the utilization of the aisle, so we can get :

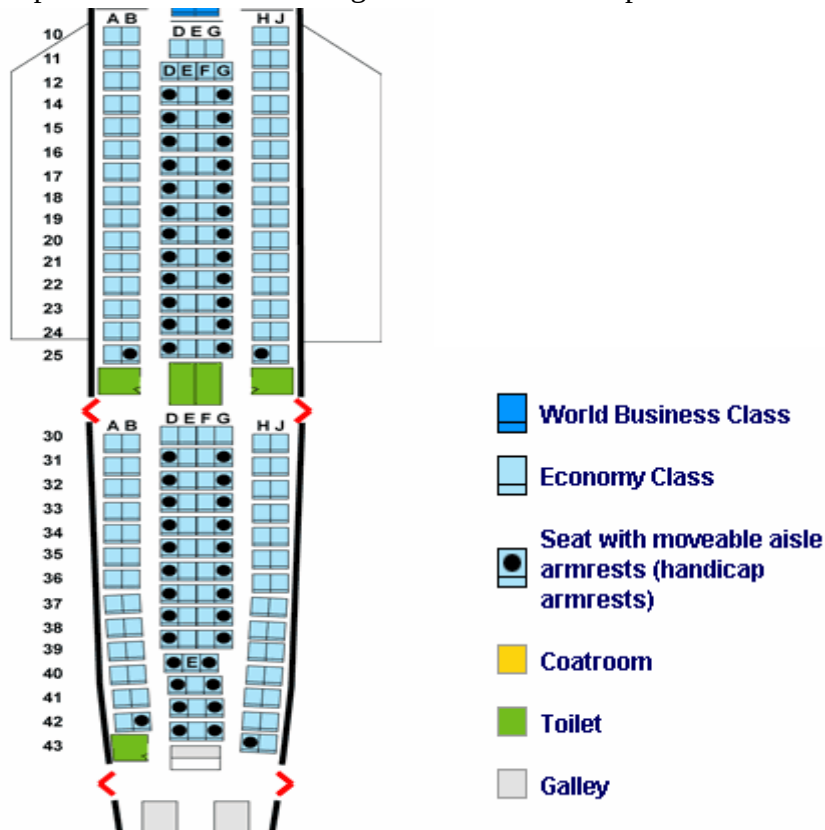
$$u = \frac{\text{the amount of steps for stowing}}{\text{Expectation step number}}$$

As the result, we can see that the simulation model and the step model have the same conclusion.

◆ Example

We have discussed the boarding procedure about small plane, now we would use our model to solve the boarding problem of the midsize and large plane.

Midsize plane: we take the Boeing A330-200 for example



We can see from the picture that a midsize aircraft has two aisles and the seats is shape 2-4-2. And two seats in the left hand are called A and B according to the traditional sequence-from left to right. The four seats which is in middle of the airplane are respectively called D, E, F and G. The two seats which is in the right hand are denoted by H and J. (The assignment is showed in Fig.8 as above.) The aisle in the left hand is represented by Aisle One while the right aisle is denoted by Aisle Two. Because the rows which is provided with the shape 2-3-2 is in the minority, then we present the hypothesis that every row has the same as shape 2-4-2 from which we can simplify the solution. The other fact is that the passenger who is seated in Line D and E must stow his luggage in Aisle One. The same situation is existed in Line F and G. So based on the hypothesis and the fact, we divide the cabin of the airplane into two parts. We assume the Line A, B, D and E as Part One and the rest as Part Two. Because Part One and Part Two has the same situation, we can only consider one part and the other part the same. Part One(or Part Two) is deemed as the airplane cabin with only

one aisle in it. Draw on the same method as the model one (showed in Fig.9), then the result is as followed:

	L-Shaped(two aisles)	
	RI	SI
Average boarding time(simulate 100 times)	60.83	0
cost (minutes)	20.813	

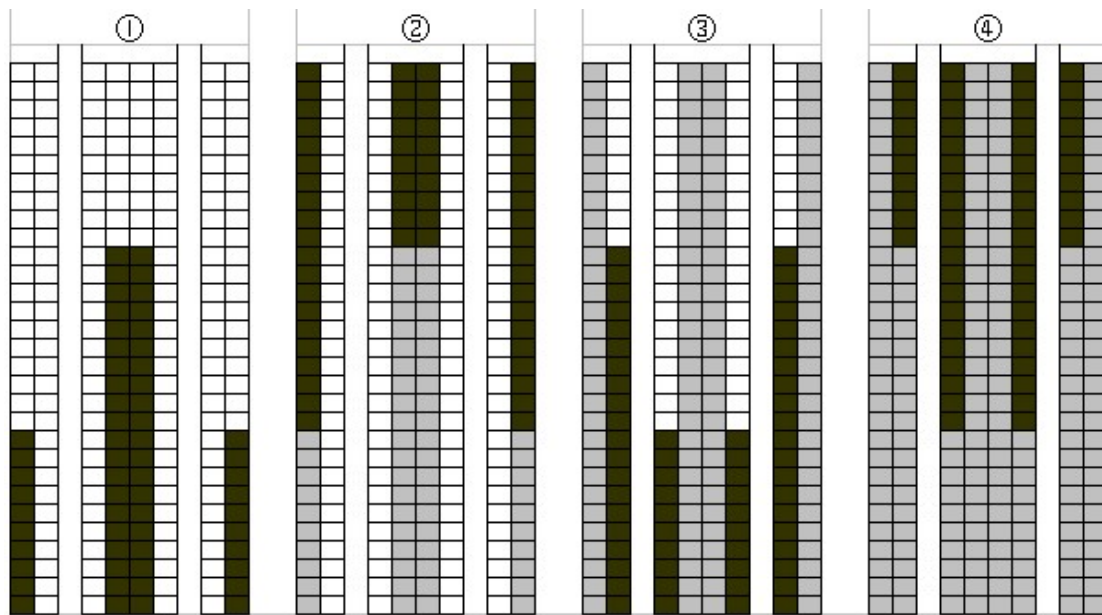


Fig.9

Large plane: In general, large plane has two stairs. Every stair is the shade of the midsize plane. We take Airbus A380 for example. The seat's configuration is showed in Fig.10.

A380

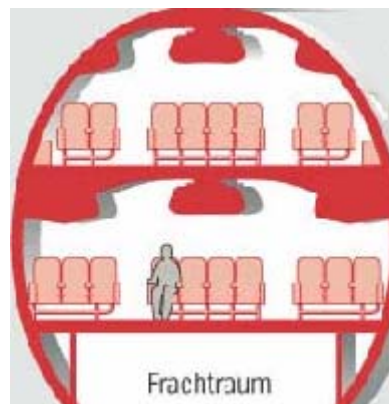


Fig.10

Since the upper floor has been discussed in midsize plane, we only analyze the under

floor.

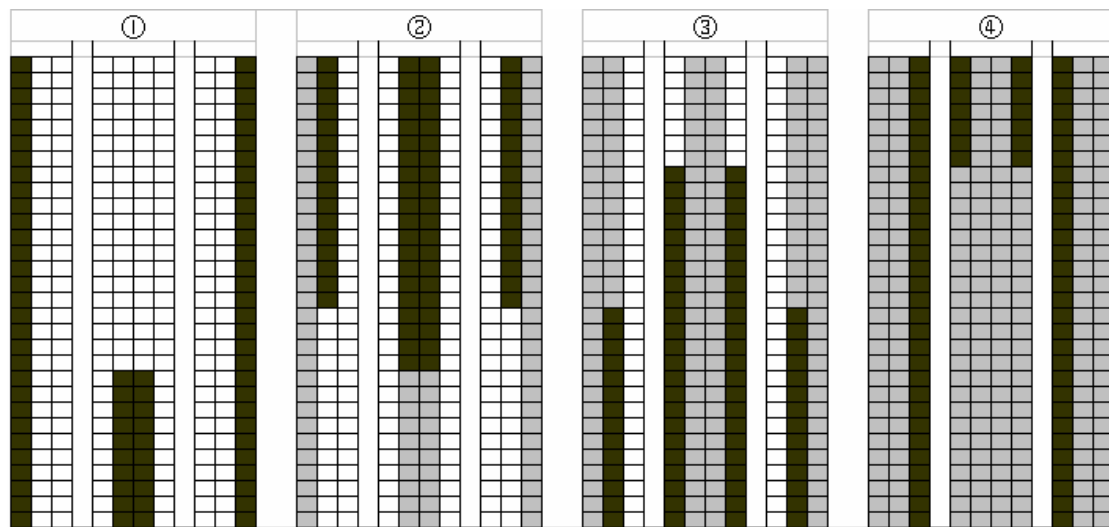


Fig.11

The shortest turnaround time is the shorter of the finish time of two stairs.

6 Deboarding

We consider the deboarding model with the small plane. The passengers deboard when the plane land. Considering the utilization of the aisle, it had better to let the passengers unload their luggage beforehand. We assume the passenger is self-organized i.e. the passenger may move to the aisle from his seat when there is not any competitor. If it is assumed that the time for unloading is equal, we may get:

116	70	24		1	47	93
117	71	25		2	48	94
118	72	26		3	49	95
119	73	27		4	50	96
120	74	28		5	51	97
121	75	29		6	52	98
122	76	30		7	53	99
123	77	31		8	54	100
124	78	32		9	55	101
125	79	33		10	56	102
126	80	34		11	57	103
127	81	35		12	58	104
128	82	36		13	59	105
129	83	37		14	60	106
130	84	38		15	61	107
131	85	39		16	62	108
132	86	40		17	63	109
133	87	41		18	64	110
134	88	42		19	65	111
135	89	43		20	66	112
136	90	44		21	67	113
137	91	45		22	68	114
138	92	46		23	69	115
Notice: The number denote the order of deboarding						

7 Improving the model

Tolerance of error Model

In this model, we consider the probability of the situation that someone disobey the rule which is presented in corresponding strategy. We want to measure what will happen under the wrong behavior and how. The processes are as bellows:

- (1) We simulate the wrong behavior and calculate the gap of the average boarding time between no wrong behavior situation and wrong behavior situation.
- (2) Compare the gap on each strategy and the gap is the measurable value of the tolerance of error.

The ‘no wrong behavior’ situation’s average boarding time is showed in Table 3. The ‘wrong behavior’ situation will be showed as follows:

	ROTATING-ZONE		BACK-TO-FRONT		OUTSIDE-IN		REVERSE-PYRAMID		Improvement for 'outside-in'		L-shaped sequence system	
	RI	SI	RI	SI	RI	SI	RI	SI	RI	SI	RI	SI
Average boarding time with a wrong behavior(simulate 100 times)	80.77	68.55	80.34	69.25	70.53	1.72	70.49	1.46	68.33	1.49	69.86	1.74
Gap interferences	0.01	0.78	1.79	-0.39	0.4	1.72	-0.3	1.46	0.19	1.49	0.4	1.74
cost (minutes)	25.461519		25.363448		23.04256		23.033113		22.537		22.8887	
Gap (minutes)	0.061519		0.463448		0.142561		-0.06689		0.04699		0.129682	

Conclusion: The result shows that the best strategy according to the seat tolerance of error of all these six strategies is back-to-front. And for the aisle is reverse-pyramid.

But the weight for the seat tolerance of error is much smaller than the aisle tolerance of error. So the strategy back-to-front with the biggest aisle tolerance of error and the smallest seat tolerance of error has the change to have the most gap time.

(1) The sequence for the seat tolerance of error is back-to-front >rotating-zone>reverse-pyramid>improvement of outside-in>outside-in> L-shaped sequence.

(2) The sequence for the aisle tolerance of error is reverse-pyramid>rotating-zone>improvement of outside-in >outside-in= L-shaped sequence>back-to-front.

(3) The sequence for the total tolerance of error is reverse-pyramid> improvement of outside-in> rotating-zone> L-shaped sequence> outside-in>back-to-front.

8 Strengths and Weaknesses

Strengths

- We use two methods to measure the effect for each strategy according to different objects, and get the same result that approximate the actual situation. So we believe our result has practical value.
- We apply comprehensive analysis in our model design. Given an actual plane, we can simulate a good procedure.

Weakness

- Our assumption may be simple, so once the emergency happens, the result may be different

9 References

- [1] van den Briel, M., Villalobos, J., and Hogg, G. (2003). The Aircraft Boarding Problem, Proc. Of the 12'th Industrial Eng. Res. Conf., IERC, CD ROM only, paper nr. 2153.
- [2] van Landeghem, H., and Beuselinck, A. (2002). Reducing passenger boarding time in airplanes: A simulation approach, European J. of Operations Research 142 , 294–308.
- [3] Marelli, S., Mattocks, G., and Merry, R. (2000). The role of computer simulation in reducing airplane turn time, Boeing Aero Magazine, Issue 1.
- [4] H. Van Landeghem, A. Beuselinck / European Journal of Operational Research 142 (2002) 294 – 308.
- [5] Menkes H. L. van den Briel, J. Rene Villalobos, Gary L. Hogg. The Aircraft Boarding Problem: Department of Industrial Engineering Arizona State University

10 Appendixes:

Program 1(simulation for back to front)

p=138;

M=zeros(1,100);

N=zeros(1,100);

W=zeros(1,100);

for q=1:100

 k=0;

 s=0;

A=[randperm(30)+108*ones(1,30),randperm(30)+78*ones(1,30),randperm(30)+48*ones(1,30),randperm(24)+24*ones(1,24),randperm(24)];

 for i=1:p-1

 if (A(1,i)>A(1,i+1))&(floor((A(1,i)-1)/6)~=floor((A(1,i+1)-1)/6))

 k;

 else

 k=k+1;

 end

 end

 for i=2:p

 if mod(A(1,i),3)==2

 for j=1:i-1

 if A(1,j)==A(1,i)-1

 s=s+1;

 end

 end

 end

```

        if mod(A(1,i),3)==0
            for j=1:i-1
                if (A(1,j)==A(1,i)-1)|(A(1,j)==A(1,i)-2)
                    s=s+1;
                end
            end
        end
    end
end
M(1,q)=k;
N(1,q)=s;
W(1,q)=0.6*k+0.4*s;
end
M'
N'
W'

```

Program 2(simulation for rotating zone system)

```

p=138;
M=zeros(1,100);
N=zeros(1,100);
W=zeros(1,100);
for q=1:100
    k=0;
    s=0;

    A=[randperm(30)+108*ones(1,30),randperm(30),randperm(30)+78*ones(1,30),randperm(24)+30*
ones(1,24),randperm(24)+54*ones(1,24)];
    for i=1:p-1
        if (A(1,i)>A(1,i+1))&(floor((A(1,i)-1)/6)~=floor((A(1,i+1)-1)/6))
            k;
        else
            k=k+1;
        end
    end
end
for i=2:p
    if mod(A(1,i),3)==2
        for j=1:i-1
            if A(1,j)==A(1,i)-1
                s=s+1;
            end
        end
    end
end
if mod(A(1,i),3)==0
    for j=1:i-1

```

```

        if (A(1,j)==A(1,i)-1)|(A(1,j)==A(1,i)-2)
            s=s+1;
        end
    end
end
end
M(1,q)=k;
N(1,q)=s;
W(1,q)=0.6*k+0.4*s;
end
M'
N'
W'

```

Program 3(simulation for outside-in)

```

p=138;
M=zeros(1,100);
N=zeros(1,100);
W=zeros(1,100);
for q=1:100
    k=0;
    s=0;
    A=[3*randperm(46),3*randperm(46)-ones(1,46),3*randperm(46)-2*ones(1,46)];
    for i=1:p-1
        if (A(1,i)>A(1,i+1))&(floor((A(1,i)-1)/6)~=floor((A(1,i+1)-1)/6))
            k;
        else
            k=k+1;
        end
    end
end
for i=2:p
    if mod(A(1,i),3)==2
        for j=1:i-1
            if A(1,j)==A(1,i)-1
                s=s+1;
            end
        end
    end
    if mod(A(1,i),3)==0
        for j=1:i-1
            if (A(1,j)==A(1,i)-1)|(A(1,j)==A(1,i)-2)
                s=s+1;
            end
        end
    end
end

```

```

        end
    end
end
M(1,q)=k;
N(1,q)=s;
W(1,q)=0.6*k+0.4*s;
end
M'
N'
W'

```

Program 4(simulation for improvement of outside-in)

```

p=138;
M=zeros(1,100);
N=zeros(1,100);
W=zeros(1,100);
for q=1:100
    k=0;
    s=0;

A=[3*(2*randperm(23)-ones(1,23)),6*randperm(23),6*randperm(23)-4*ones(1,23),6*randperm(23)-ones(1,23),6*randperm(23)-5*ones(1,23),6*randperm(23)-2*ones(1,23)];
    for i=1:p-1
        if (A(1,i)>A(1,i+1))&(floor((A(1,i)-1)/6)~=floor((A(1,i+1)-1)/6))
            k;
        else
            k=k+1;
        end
    end
end
for i=2:p
    if mod(A(1,i),3)==2
        for j=1:i-1
            if A(1,j)==A(1,i)-1
                s=s+1;
            end
        end
    end
end
if mod(A(1,i),3)==0
    for j=1:i-1
        if (A(1,j)==A(1,i)-1)|(A(1,j)==A(1,i)-2)
            s=s+1;
        end
    end
end
end
end

```

```
end
M(1,q)=k;
N(1,q)=s;
W(1,q)=0.6*k+0.4*s;
end
M'
N'
W'
```

Program 5(simulation for L-shaped)

```
p=138;
M=zeros(1,100);
N=zeros(1,100);
W=zeros(1,100);
for q=1:100
    k=0;
    s=0;
    B=6*randperm(35);
    for i=1:35
        if B(1,i)>140
            B(1,i)=B(1,i)-75;
        end
    end
    C=6*randperm(34)-ones(1,34);
    for i=1:34
        if C(1,i)>140
            C(1,i)=C(1,i)-140;
        end
    end
    D=6*randperm(35)-4*ones(1,35);
    for i=1:35
        if D(1,i)>138
            D(1,i)=D(1,i)-70;
        end
    end
    E=6*randperm(34)-5*ones(1,34);
    for i=1:34
        if E(1,i)>138
            E(1,i)=E(1,i)-135;
        end
    end
    A=[B,C,D,E];
```

```

for i=1:p-1
    if (A(1,i)>A(1,i+1))&(floor((A(1,i)-1)/6)~=floor((A(1,i+1)-1)/6))
        k;
    else
        k=k+1;
    end
end
for i=2:p
    if mod(A(1,i),3)==2
        for j=1:i-1
            if A(1,j)==A(1,i)-1
                s=s+1;
            end
        end
    end
    if mod(A(1,i),3)==0
        for j=1:i-1
            if (A(1,j)==A(1,i)-1)|(A(1,j)==A(1,i)-2)
                s=s+1;
            end
        end
    end
end
M(1,q)=k;
N(1,q)=s;
W(1,q)=0.6*k+0.4*s;
end
M'
N'
W'

```

Program 6(simulation for the reverse pyramid)

```

p=138;
M=zeros(1,100);
N=zeros(1,100);
W=zeros(1,100);
for q=1:100
    k=0;
    s=0;
    B=3*randperm(28)+53*ones(1,28);
    for i=1:28
        if B(1,i)<85

```

```

        B(1,i)=B(1,i)-29;
    end
end
C=3*randperm(28)-ones(1,28);
for i=1:28
    if C(1,i)<25
        C(1,i)=C(1,i)+1;
    end
end
D=3*randperm(28)+52*ones(1,28);
for i=1:28
    if D(1,i)<77
        D(1,i)=D(1,i)-53;
    end
end
A=[3*randperm(28)+54*ones(1,28),B,C,D,3*randperm(26)-2*ones(1,26)];
for i=1:p-1
    if (A(1,i)>A(1,i+1))&(floor((A(1,i)-1)/6)~=floor((A(1,i+1)-1)/6))
        k;
    else
        k=k+1;
    end
end
for i=2:p
    if mod(A(1,i),3)==2
        for j=1:i-1
            if A(1,j)==A(1,i)-1
                s=s+1;
            end
        end
    end
    if mod(A(1,i),3)==0
        for j=1:i-1
            if (A(1,j)==A(1,i)-1)|(A(1,j)==A(1,i)-2)
                s=s+1;
            end
        end
    end
end
M(1,q)=k;
N(1,q)=s;
W(1,q)=0.6*k+0.4*s;
end
M'
```

N'

W'

simu lati on	ROTATING -ZONE	BACK-TO-F RONT	OUTSIDE-IN	REVERS E-PYRA MID	Improvem ent of out-side -in	l-shaped
1	74.6	76.4	40.2	42.6	41.4	42
2	74.4	72	45.6	44.4	41.4	41.4
3	75.6	73.4	42	42	40.2	44.4
4	76.8	76	40.2	42	39	46.8
5	74.8	72.6	43.2	40.2	39.6	43.2
6	81.2	77.8	44.4	40.8	37.8	42
7	75.4	76.6	40.8	48.6	40.2	42.6
8	78	77.8	40.2	42.6	42.6	42.6
9	75.4	73.6	40.8	40.8	41.4	40.8
10	75.6	74.8	44.4	39.6	42.6	40.2
11	77.2	72.8	43.8	41.4	41.4	40.8
12	78.6	74.4	38.4	42	41.4	42
13	73.8	75.6	43.8	42.6	42	41.4
14	76	75.4	41.4	42	43.2	40.8
15	72.2	75	43.8	44.4	43.2	43.8
16	71.8	70.2	39.6	42	39.6	45.6
17	71.6	74.6	40.2	43.8	42	41.4
18	75	73.2	43.8	43.2	42	42
19	76.4	70.4	37.8	41.4	42	39
20	74	73.8	44.4	38.4	40.2	39.6
21	80.2	74	39	41.4	43.2	43.2
22	80.4	75.6	43.8	42.6	44.4	39.6
23	79.8	78	42	42.6	39	41.4
24	71.4	78	42.6	40.2	40.2	42
25	76.8	75.2	41.4	43.2	39.6	37.8
26	77	73.4	39.6	43.2	42	43.2
27	76.2	71.4	42	41.4	38.4	38.4
28	73.4	72.8	37.8	40.2	37.8	42
29	78.4	77.2	45.6	41.4	42	39.6
30	79.6	77.6	40.8	42.6	40.8	42
31	78.4	71.2	43.2	44.4	40.8	43.8
32	71.6	69.8	40.2	47.4	41.4	40.8
33	70.2	72.6	42.6	41.4	40.8	44.4
34	78.4	79.6	42	40.2	39.6	44.4

35	75	73.2	43.2	39.6	43.8	39.6
36	78	73.8	42	45.6	40.2	39.6
37	77.6	71.4	42	43.2	44.4	42
38	75.6	75.8	41.4	40.2	40.2	40.8
39	71.4	78.6	42	42	42	42.6
40	71.2	74.8	42.6	39	40.8	45
41	66.6	71.6	42	44.4	37.8	42.6
42	79.2	79.6	45	39.6	43.2	40.2
43	78.2	73.2	41.4	44.4	40.2	41.4
44	78	76.8	41.4	39.6	39.6	40.8
45	72.6	78.6	42	43.8	40.2	44.4
46	81.6	70.6	40.8	42.6	39	36.6
47	78.4	80.6	40.2	42	39.6	39.6
48	78.2	73.2	43.8	46.2	39	40.8
49	79.6	74.8	41.4	46.2	40.2	42.6
50	73.2	70.8	43.8	41.4	41.4	43.2
51	81.6	75	44.4	43.8	43.2	39
52	78.6	72.8	41.4	40.2	40.8	45
53	72.6	70.2	41.4	42.6	40.2	43.8
54	77.2	74.6	41.4	42.6	40.8	43.2
55	76.4	74.4	46.8	43.2	43.2	42
56	78	76.2	40.8	41.4	40.8	39.6
57	77.6	72.8	42	45	41.4	42
58	72.4	74.4	40.8	43.2	38.4	41.4
59	74.2	74	40.2	41.4	40.2	37.8
60	76.6	75.8	40.2	39.6	40.2	42.6
61	76.8	78.4	40.8	43.2	40.2	42
62	79.6	71	46.2	42.6	44.4	42
63	82.6	71.8	42.6	41.4	41.4	45.6
64	76.6	72.2	41.4	40.8	39.6	42
65	78.4	76.2	39	46.2	41.4	40.8
66	76.4	71	43.2	39.6	43.2	42
67	69.2	76.6	43.8	42	36.6	42
68	74.8	78.6	41.4	39.6	40.2	40.8
69	77.4	77	42	43.2	41.4	39.6
70	72	76.2	42.6	40.8	40.2	40.2
71	74.4	79	43.2	41.4	40.2	42
72	72.4	81.6	43.2	40.8	40.2	40.8
73	76.8	77.4	43.2	42.6	40.8	41.4
74	78	73	42.6	45	43.8	40.8

75	79.6	72.4	42.6	43.8	47.4	40.2
76	78.8	77.8	42	39	43.8	41.4
77	75.2	71.8	41.4	40.8	39	42.6
78	75.2	79.6	43.2	40.2	42.6	40.8
79	79	77.6	38.4	42.6	39.6	44.4
80	73.2	78.6	40.8	42.6	39.6	46.2
81	74.4	78	44.4	43.8	42	42.6
82	76.6	72.6	45.6	42	43.2	41.4
83	71.8	74.4	40.8	42	43.2	43.8
84	82.6	74.4	43.8	40.8	43.8	39.6
85	81.4	72.4	39.6	42.6	40.2	42.6
86	72	76	42	39.6	42.6	40.2
87	82.4	73.4	43.2	43.8	40.2	37.2
88	77	74.8	42	40.8	43.8	42.6
89	73	75	42	42	39	45
90	69.2	74.8	42.6	42	40.2	42.6
91	82.8	82.2	43.2	45	45.6	40.8
92	75.8	74.6	43.8	40.8	37.8	42
93	66	73.6	42	45	44.4	40.2
94	70	72.6	44.4	44.4	42	41.4
95	79.8	77.8	42.6	40.8	41.4	39.6
96	77.6	74.2	42.6	43.2	42	37.8
97	76.8	76.2	39	43.8	43.2	46.2
98	73.8	74	40.8	40.2	41.4	43.8
99	75.8	78.8	40.8	43.8	41.4	41.4
100	71.2	74.4	40.8	38.4	43.2	44.4
平均值	75.962	74.924	42.054	42.228	41.202	41.76

To code the seat

132	133	134	135	136	137
126	127	128	129	130	131
120	121	122	123	124	125
114	115	116	117	118	119
108	109	110	111	112	113
102	103	104	105	106	107
96	97	98	99	100	101
90	91	92	93	94	95
84	85	86	87	88	89
78	79	80	81	82	83
72	73	74	75	76	77
66	67	68	69	70	71
60	61	62	63	64	65
54	55	56	57	58	59
48	49	50	51	52	53
42	43	44	45	46	47
36	37	38	39	40	41
30	31	32	33	34	35
24	25	26	27	28	29
18	19	20	21	22	23
12	13	14	15	16	17
6	7	8	9	10	11
0	1	2	3	4	5

The game algorithms:

```

#include<stdio.h>
#include<math.h>
#include<stdlib.h>
#include<string.h>
#define N 138
void main()
{
    int e[23][6]={0};
    int i,j=0,k,flag[23][2]={0},l,d=0,q;
    FILE *fp;
    struct f
    {
        int x;
        int y;
    }a[N];
    int p,sum=0;
    struct
    {
        int m,n;
        struct f c;
    }b[23];
    fp=fopen("c:\\a.txt","w");
    for(q=0;q<10;q++)

```

```
{
    for(i=0;i<23;i++)
    {
        b[i].m=0;
        b[i].n=10;
        b[i].c.x=24;
    }
    for(i=0;i<23;i++)
        for(j=0;j<6;j++)
            e[i][j]=0;

    for(i=0;i<N;i++)
    {
        scanf("%d %d",&a[i].x,&a[i].y);
    }
    for(i=0;i<N;i++)
    {
        printf("%d %d\n",a[i].x,a[i].y);
    }

    for(k=0;;k++)
    {

        for(i=0;i<23;i++)
        {
            if(b[i].c.x==i&&(b[i].m==1||b[i].m==2))
            {
                b[i].m=2;
                if(b[i].n==10)
                    b[i].n=2;
                else
                    if(b[i].n!=0)
                        (b[i].n)--;
                else
                {
                    b[i].n=10;
                    b[i].m=3;
                    continue;
                }
            }
            if(b[i].m==3)
            {
                p=b[i].c.y;
```

```
        if(p==1||p==2||p==0)
        {

if(flag[i][0]==0&&((p==1&&e[i][2]==1)||(p==0&&(e[i][1]==1||e[i][2]==1))))
        {
            flag[i][0]=1;
        }
        else
        {b[i].m=0;
            flag[i][0]=0;
            e[i][p]=1;

        }
    }
    else
    {

if(flag[i][1]==0&&((p==4&&e[i][3]==1)||(p==5&&(e[i][4]==1||e[i][3]==1))))
        {
            flag[i][1]=1;
        }
        else
        {
            b[i].m=0;
            flag[i][1]=0;
            e[i][p]=1;

        }
    }
}

}

for(i=0;i<22;i++)
{
    if(b[i+1].m==1&&b[i].m==0)
    {
        b[i].c.x=b[i+1].c.x;
        b[i].c.y=b[i+1].c.y;
        b[i+1].m=0;
        b[i].m=1;

    }
}
```

```

    }
    if(b[22].m==0&& j<N)
    {
        b[22].c.x=a[j].x;
        b[22].c.y=a[j].y;
        b[22].m=1;
        j++;
    }

    for(l=0;l<23;l++)
        if( b[l].m==0)
            d++;
    if(d!=23)
    {
        d=0;
    }
    else
        break;

}
fprintf(fp,"%d\n",k);
sum+=k;
}
fprintf(fp,"%d\n",sum/10);
fclose(fp);
}

```

The different coding methods:

The outside-in:

```

M=zeros(276,10);
N=zeros(2760,1);
for i=1:10
    B=6*randperm(46)-ones(1,46);
    for t=1:46
        if B(1,t)>140
            B(1,t)=B(1,t)-143;
        end
    end
    C=6*randperm(46)-2*ones(1,46);
    for t=1:46
        if C(1,t)>140

```

```

        C(1,t)=C(1,t)-141;
    end
end
D=6*randperm(46)-3*ones(1,46);
for t=1:46
    if D(1,t)>140
        D(1,t)=D(1,t)-139;
    end
end
A=[B,C,D];
for j=1:138
    M(2*j-1,i)=floor(A(1,j)/6);
    M(2*j,i)=mod(A(1,j),6);
end
end
for n=1:10
    for m=1:276
        N(276*(n-1)+m,1)=M(m,n);
    end
end
N

```

The V-sharped

```

M=zeros(276,10);
N=zeros(2760,1);
for i=1:10
    B=6*randperm(28)-ones(1,28);
    for t=1:28
        if B(1,t)>85
            B(1,t)=B(1,t)-89;
        end
    end
    C=6*randperm(28)-2*ones(1,28);
    for t=1:28
        if (C(1,t)>55)&(C(1,t)<108)
            C(1,t)=C(1,t)-57;
        end
        if (C(1,t)>108)&(C(1,t)<138)
            C(1,t)=C(1,t)-28;
        end
        if (C(1,t)>138)&(C(1,t)<170)
            C(1,t)=C(1,t)-53;
        end
    end
end

```

```

D=6*randperm(28)+52*ones(1,28);
for t=1:28
    if (D(1,t)>115)&(D(1,t)<175)
        D(1,t)=D(1,t)-63;
    end
    if (D(1,t)>175)&(D(1,t)<200)
        D(1,t)=D(1,t)-64;
    end
    if (D(1,t)>200)&(D(1,t)<222)
        D(1,t)=D(1,t)-83;
    end
end
E=6*randperm(28)-3*ones(1,28);
for t=1:28
    if (E(1,t)>60)&(E(1,t)<120)
        E(1,t)=E(1,t)-61;
    end
    if (E(1,t)>120)&(E(1,t)<143)
        E(1,t)=E(1,t)-8;
    end
    if (E(1,t)>143)&(E(1,t)<170)
        E(1,t)=E(1,t)-29;
    end
end
F=6*randperm(26)+57*ones(1,26);
for t=1:26
    if F(1,t)>140
        F(1,t)=F(1,t)-79;
    end
end
A=[B,C,D,E,F];
for j=1:138
    M(2*j-1,i)=floor(A(1,j)/6);
    M(2*j,i)=mod(A(1,j),6);
end
end
for n=1:10
    for m=1:276
        N(276*(n-1)+m,1)=M(m,n);
    end
end
N

```

Back to front:

```

M=zeros(276,10);
N=zeros(2760,1);
for i=1:10

A=[randperm(30)-ones(1,30),randperm(30)+29*ones(1,30),randperm(30)+59*ones(1,30),randperm(24)+89*ones(1,24),randperm(24)+113*ones(1,24)];
    for j=1:138
        M(2*j-1,i)=floor(A(1,j)/6);
        M(2*j,i)=mod(A(1,j),6);
    end
end
for n=1:10
    for m=1:276
        N(276*(n-1)+m,1)=M(m,n);
    end
end
N

```

The rotating zone system:

```

M=zeros(276,10);
N=zeros(2760,1);
for i=1:10

A=[randperm(30)-ones(1,30),randperm(30)+107*ones(1,30),randperm(30)+29*ones(1,30),randperm(24)+83*ones(1,24),randperm(24)+59*ones(1,24)];
    for j=1:138
        M(2*j-1,i)=floor(A(1,j)/6);
        M(2*j,i)=mod(A(1,j),6);
    end
end
for n=1:10
    for m=1:276
        N(276*(n-1)+m,1)=M(m,n);
    end
end
N

```